# CARD 01 — PROPANE

Molecular Formula: $C_3H_8$

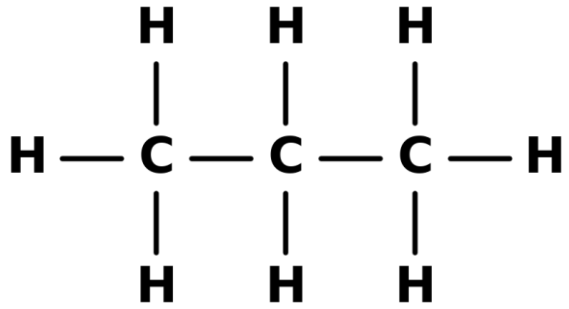
## 1. MOLECULAR FORMULA

$$C_3H_8$$

**Propane**

11 atoms (3 C, 8 H)

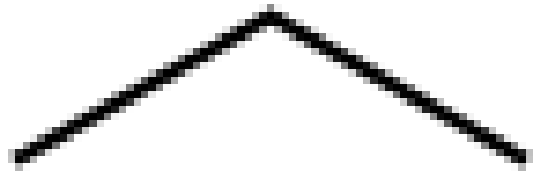
## 2. LEWIS STRUCTURE




## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2CH_3$$

## 4. BOND-LINE FORMULA (KETCHER)



**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 02 — ETHANOL

Molecular Formula: $C_2H_6O$

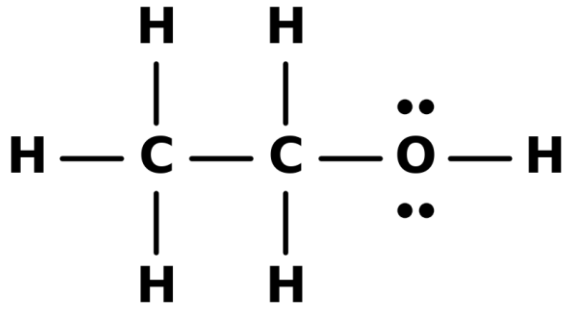
## 1. MOLECULAR FORMULA

$$C_2H_6O$$

**Ethanol**

9 atoms (2 C, 6 H, 1 O)

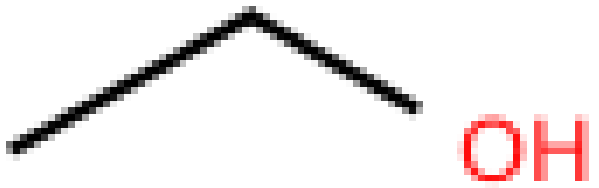
## 2. LEWIS STRUCTURE




## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CH_2OH$$

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 03 — ACETONE

Molecular Formula: $C_3H_6O$

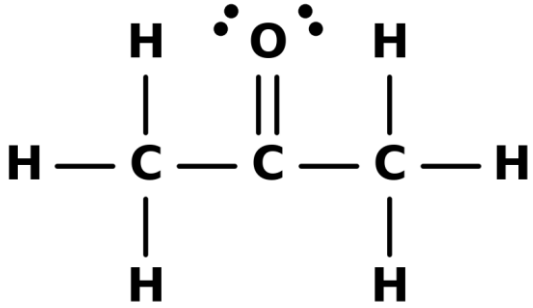
## 1. MOLECULAR FORMULA

$$C_3H_6O$$

**Acetone (propan-2-one)**

10 atoms (3 C, 6 H, 1 O)

## 2. LEWIS STRUCTURE

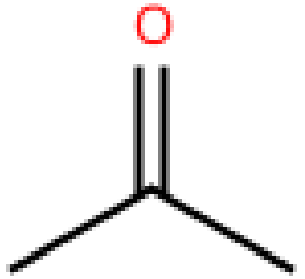



## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COCH_3$$

also written $(CH_3)_2C{=}O$

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 04 — ACETIC ACID

Molecular Formula: $C_2H_4O_2$

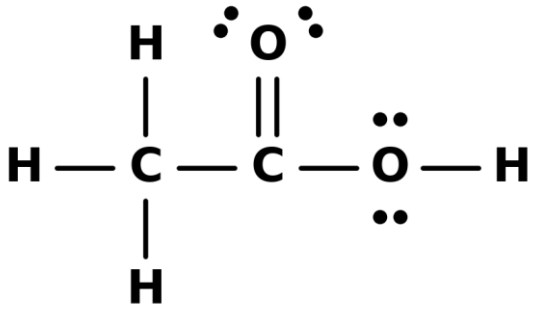
## 1. MOLECULAR FORMULA

$$C_2H_4O_2$$

**Acetic acid (ethanoic acid)**

8 atoms (2 C, 4 H, 2 O)

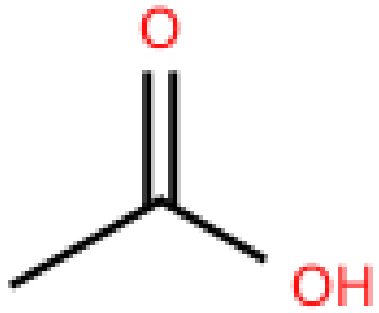
## 2. LEWIS STRUCTURE




## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOH$$

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 05 — ETHYL ACETATE

Molecular Formula: $C_4H_8O_2$

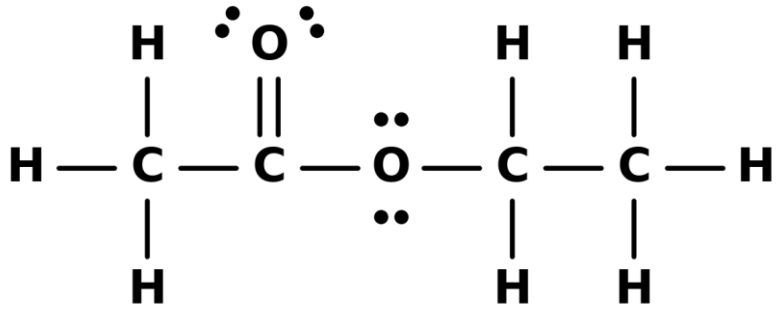
## 1. MOLECULAR FORMULA

$$C_4H_8O_2$$

**Ethyl acetate (ethyl ethanoate)**

14 atoms (4 C, 8 H, 2 O)

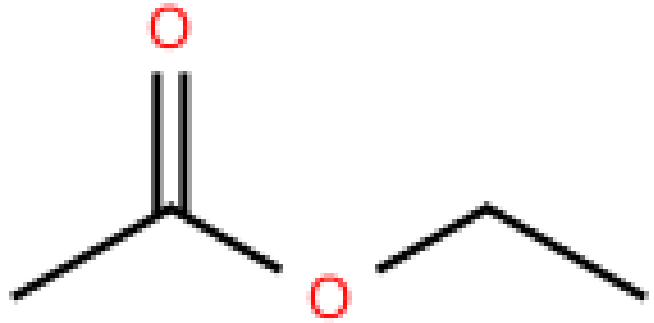
## 2. LEWIS STRUCTURE



## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COOCH_2CH_3$$

## 4. BOND-LINE FORMULA (KETCHER)



**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 06 — CYCLOHEXANE

Molecular Formula: $C_6H_{12}$

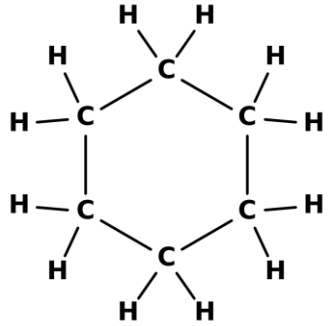
## 1. MOLECULAR FORMULA

$$C_6H_{12}$$

**Cyclohexane**

18 atoms (6 C, 12 H)

## 2. LEWIS STRUCTURE

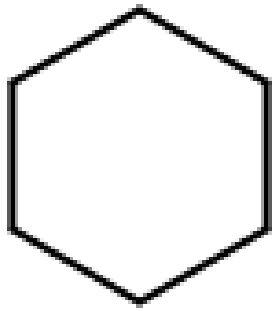



## 3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_6$$

ring of six $CH_2$ groups

## 4. BOND-LINE FORMULA (KETCHER)



**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 07 — CYCLOHEXANONE

Molecular Formula: $C_6H_{10}O$

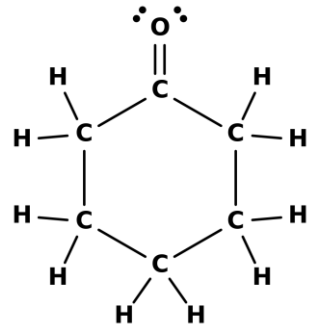
## 1. MOLECULAR FORMULA

$$C_6H_{10}O$$

**Cyclohexanone**

17 atoms (6 C, 10 H, 1 O)

## 2. LEWIS STRUCTURE

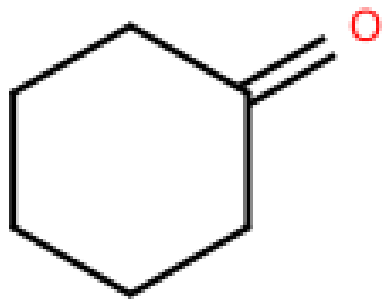



## 3. CONDENSED STRUCTURAL FORMULA

$$(CH_2)_5C{=}O$$

ring of five $CH_2$ groups and one C=O

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 08 — ACETAMINOPHEN

Molecular Formula: $C_8H_9NO_2$

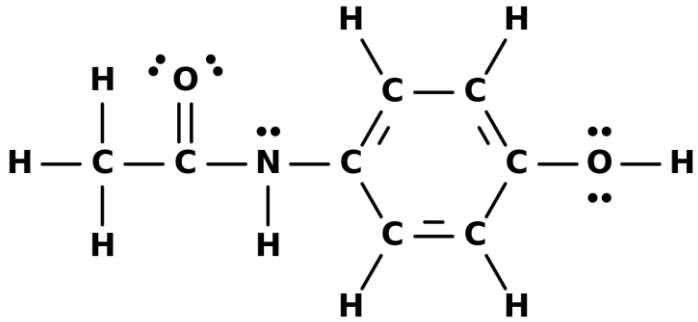
## 1. MOLECULAR FORMULA

$$C_8H_9NO_2$$

**Acetaminophen (paracetamol)**

20 atoms (8 C, 9 H, 1 N, 2 O)

## 2. LEWIS STRUCTURE

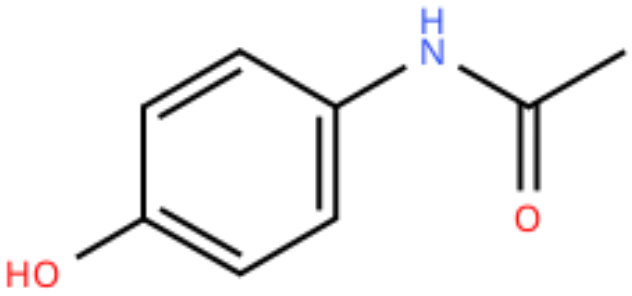


## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3CONH–C_6H_4–OH$$

para (1,4) substitution on the ring

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 09 — ASPIRIN

Molecular Formula: $C_9H_8O_4$

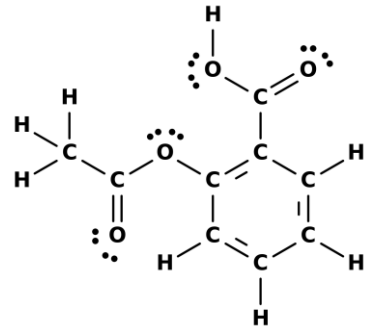
## 1. MOLECULAR FORMULA

$$C_9H_8O_4$$

**Aspirin (acetylsalicylic acid)**

21 atoms (9 C, 8 H, 4 O)

## 2. LEWIS STRUCTURE

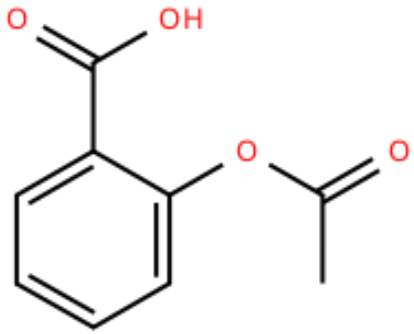


## 3. CONDENSED STRUCTURAL FORMULA

$$CH_3COO–C_6H_4–COOH$$

ortho (1,2) substitution on the ring

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**

# CARD 10 — CAFFEINE

Molecular Formula: $C_8H_{10}N_4O_2$

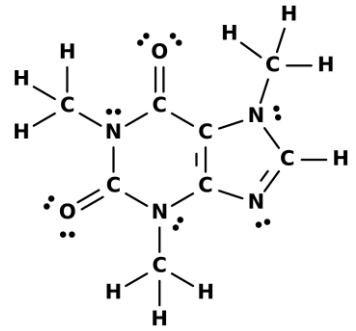
## 1. MOLECULAR FORMULA

$$C_8H_{10}N_4O_2$$

**Caffeine (1,3,7-trimethylxanthine)**

24 atoms (8 C, 10 H, 4 N, 2 O)

## 2. LEWIS STRUCTURE

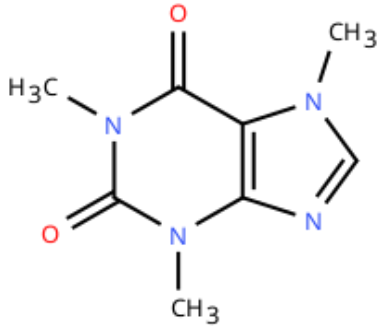


## 3. CONDENSED STRUCTURAL FORMULA

**N1($CH_3$)–C2(=O)–N3($CH_3$)–C4=C5–C6(=O)–N1**
**C4–N9=C8(H)–N7($CH_3$)–C5**

written with ring labels: six-membered ring on the first line, five-membered ring closed on the second

## 4. BOND-LINE FORMULA (KETCHER)




**atom count verified | valence checked | bond order checked | same connectivity confirmed**